Equity Market Trends: Volatility Regimes and Valuation Signals

Independent data-analysis exercise by Tharun Gonela, MSc Data Science, University of Roehampton. Python (pandas, NumPy, Matplotlib). Data: public monthly S&P 500 series (datasets/s-and-p-500 on GitHub).

Question

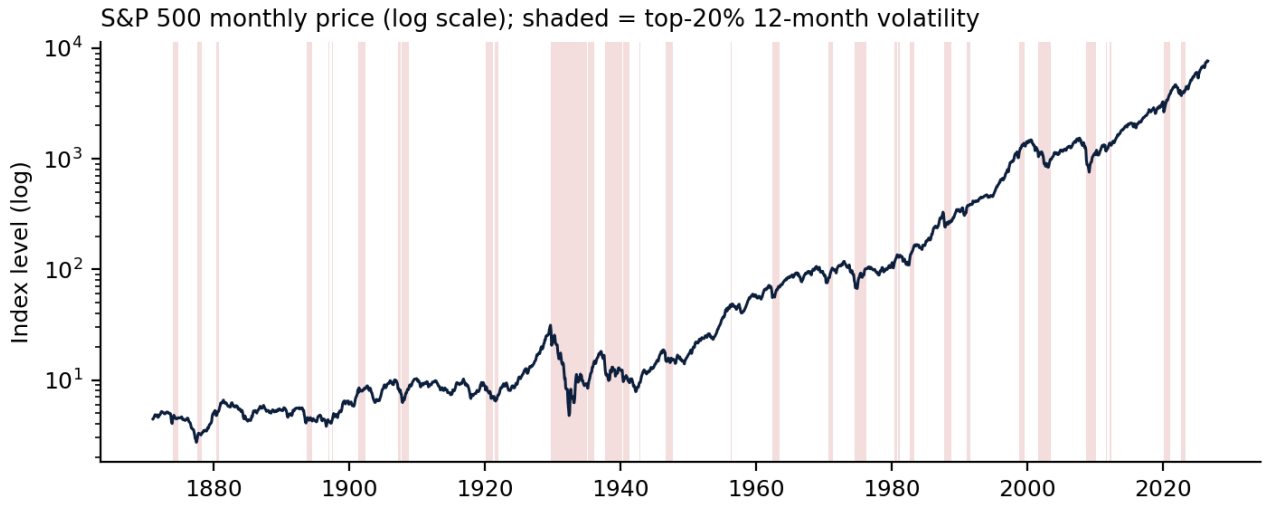
Can simple, transparent measures, namely volatility regimes and starting valuation, say anything useful about subsequent equity returns, and do they survive an honest out-of-sample test?

Data and method

Monthly index levels 1871 to September 2026. Zeros in the file are placeholders, so I treated them as missing. Real price and CAPE (cyclically adjusted P/E) are only available to September 2023, so valuation tests use data up to then. All returns are **price returns** (no dividends). I have not independently verified the post-2023 price points, and the series looks like monthly averages, which smooth volatility. Rolling windows overlap, so observations are not independent.

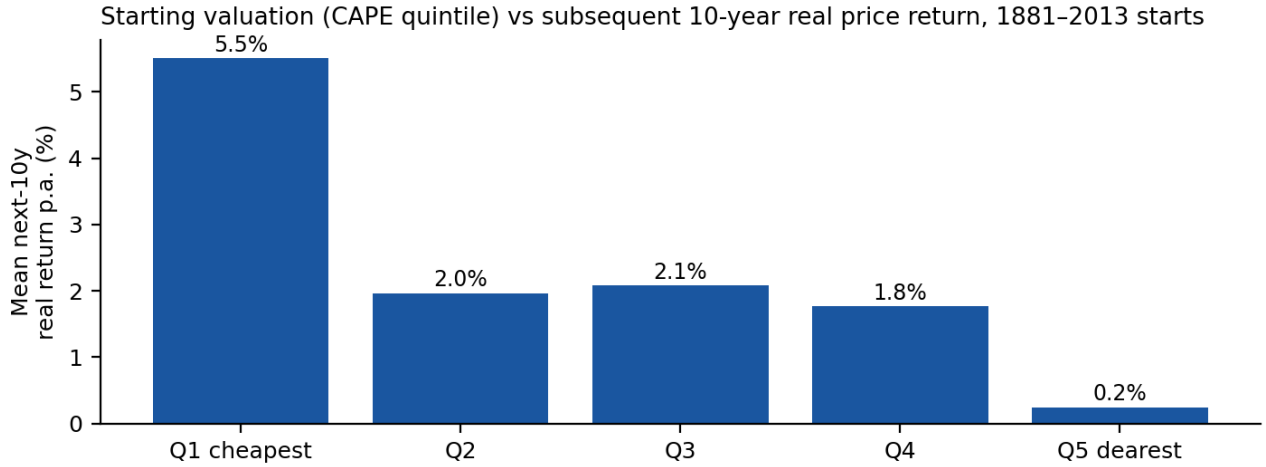
Finding 1: high-volatility months were followed by higher average returns

Months in the top 20% of rolling 12-month volatility (above about 14.5% annualised) were followed by an average 12-month price return of **8.2%**, against **6.1%** in other months (1871 to 2025). Since 1996 the gap is wider: **16.2%** versus **8.0%**. Interpretation: volatility spikes clustered around drawdowns that later recovered. This is descriptive only. I ran no significance test, the windows overlap, and it is not a trading rule.



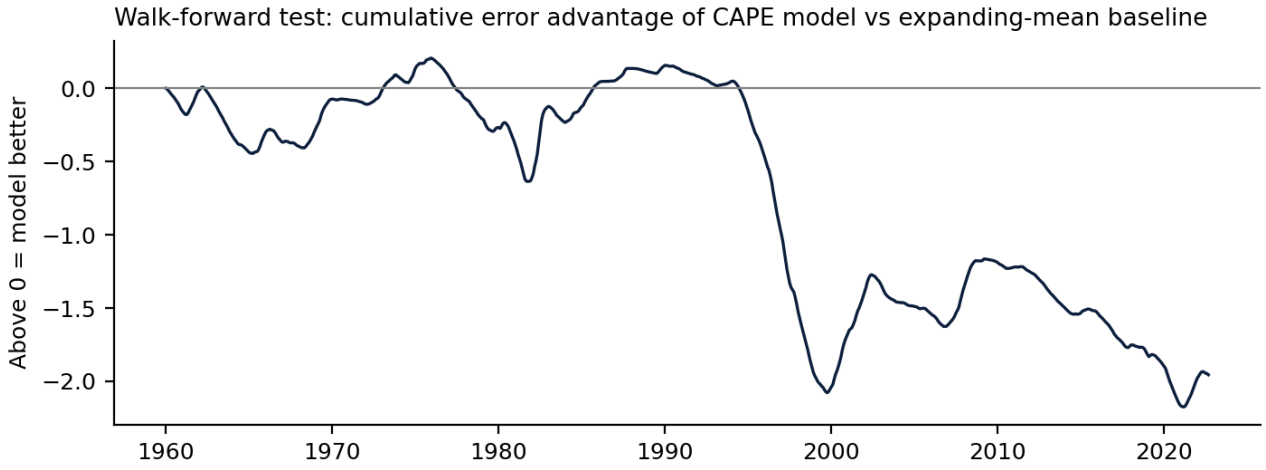
Finding 2: starting valuation relates to 10-year returns

Across 1,593 monthly starting points (1881 to 2013), CAPE and the next 10-year annualised real price return have a correlation of **-0.33**. The cheapest quintile of starting valuations averaged **5.5% a year**; the most expensive averaged **0.2% a year**. Because 10-year windows overlap heavily, the effective sample is only a little over a dozen independent decades, so this is suggestive, not proof.



Finding 3: the valuation signal did not beat a simple baseline out of sample

I tested a walk-forward model: an expanding-window regression of the next 12-month real return on log CAPE, using only outcomes already known at each date, tested from 1960 to September 2022 (753 months). It did **not** beat an expanding-mean baseline: out-of-sample R-squared was **-0.10** overall, **+0.02** for 1960 to 1989 and **-0.21** for 1990 to 2022. A relationship that appears over long horizons did not become a usable 12-month forecast, especially after valuations shifted structurally in the 1990s.



Limitations and next steps

Single index; price returns only; no transaction costs; no formal significance testing. Next steps: add total returns and interest-rate features, try regime models (for example hidden Markov), use block-bootstrap confidence intervals to handle overlapping windows, and repeat on other asset classes.

Reproducibility

All numbers come from the attached script (market_trends_analysis.py) run on the public CSV above.